

LSGI3614 Geophysical Survey of Utilities

Utilities Survey Report

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Group ■ Students

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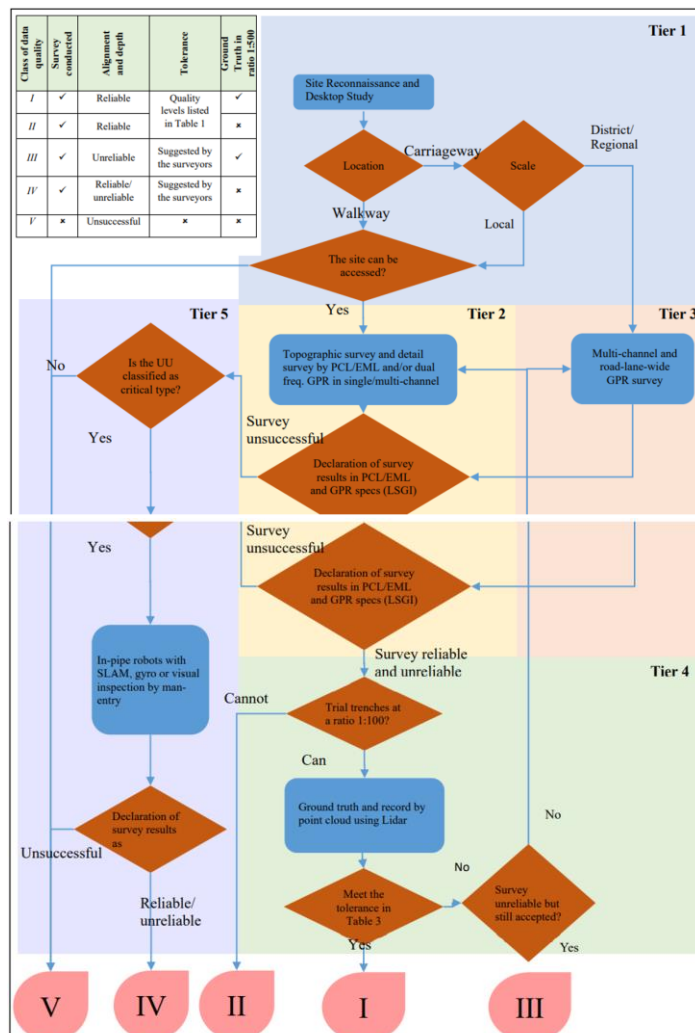
Objective

Our group was assigned to conduct an underground utility survey at Site 3, Q Leak. Q Leak was located at Liu To Road, Tsing Yi, which was operated by the Water Supplies Department for underground water mains leak detection training. This survey aims to identify the existing water mains alignment from SAV2 to GV50, using ground penetrating radar (GPR) and pipe cable locator (PCL).

For our surveying, we measure the PCL survey on 22/9/2025, and measure the GPR survey on 6/10/2025, the weather of these two days both are sunny, and the material of the survey site is dry sand. The survey area is $8\text{m} \times 5\text{m} = 40\text{m}^2$, and the entire surface is flat.

After this survey training, we are enabled to master the use of both types of equipment.

Here is 5-tier survey plan for our group survey project:

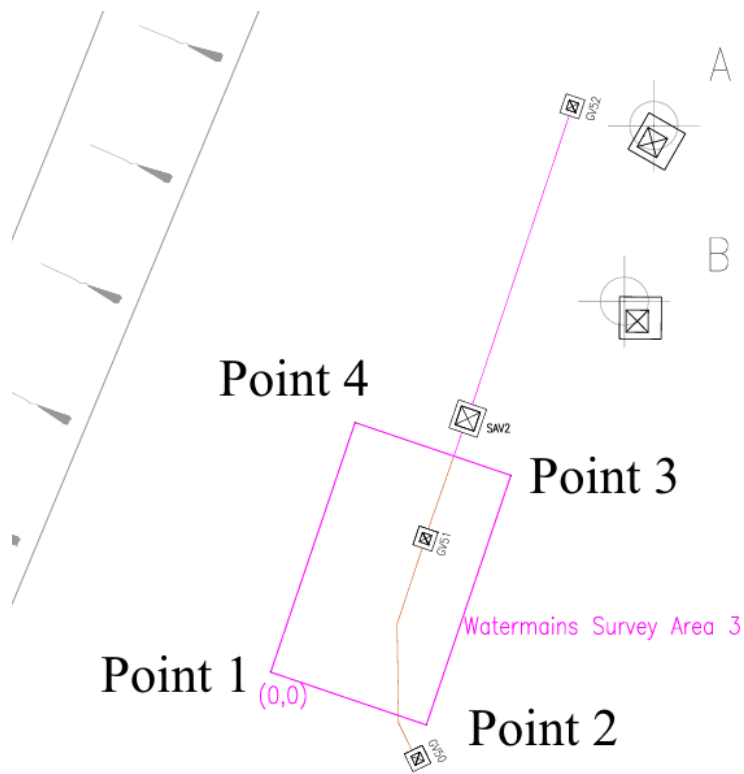


Methodology

1. Find the designated area depicted on the watermains layout plan. There were already previous markings of the exact site boundaries, thus no setting out was required.
2. Start with one valve for the PCL survey. We can use passive mode to detect any electrical cable before switching to direct connection to detect the water mains.
 - 2.1. Turn on the transmitter and set the frequency to the same as the receiver. Place the transmitter perpendicular to the alignment. The recommended starting frequency for the active signal is 33kHz.
 - 2.2. Connect the cables to the transmitter. Connect the red clamp to the valve and connect the black clamp to the grounding. (e.g. manhole)
 - 2.3. Start receiving 1.5 meters around the valve using the peak mode, and find the maximum in the low-high-low signal of the receiver, indicating the alignment of the pipe.
 - 2.4. Every 1.5 meters find the next maximum of the low-high-low signal. If the signal is absent in the direction of the pipe, indicates the pipe has turned and goes around the pipe to find the low-high-low.
3. Use GNSS RTK to record the coordinates of the pipe from the PCL survey.
4. Calculate the time window of the GPR survey with approximated material and the wavelength. Set the GPR with the time window and the horizontal grid spacing will be no more than 4 times the wavelength.
5. Start conducting the GPR survey.
 - 5.1. Use a tape measure to form right angles, forming a rectangle of the site and mark the grid according to the grid spacing.
 - 5.2. Calibrate the wheel by pushing the GPR along a measured distance with a tape measure.
 - 5.3. Conduct the GPR survey, pushing the GPR along each grid line starting from the origin, clearly marking the file names and the grid line order on the survey field sheet.
6. Post processing using Geolitix to generate B-scan views, C-scan slices.
7. Depict the coordinates on CAD drawings and compare the data collected from PCL and GPR.

Site location

The surveying site was located on the upper platform of Q-Leak. According to the watermains layout plan, the exact site boundary was depicted as follows:



Control point name	Easting	Northing
Point 1 (0, 0)	827938.166	824009.039
Point 2	827942.902	824007.434
Point 3	827945.469	824015.011
Point 4	827940.734	824016.616
A	827949.813	824025.629
B	827948.921	824020.295

Instrument models

	
RD8200 for PCL survey	IDS Hi-MOD for GPR survey

Designating the GPR surveying grid

a. Time window calculation

Formula for calculating the time window:

$$T_w > factor * \frac{2D_{max} * \sqrt{K}}{c}$$

T_w = Time window

c = speed of light in a vacuum

D_{max} = Expected maximum object depth

K = real part of permittivity

We assumed the factor was 2, depth was 3m, and the material was dry sand with $K=4$.

$$T_w > 2 * \frac{2 * 3m * \sqrt{4}}{299792458 m/s}$$

$$T_w > 8 * 10^{-8} s$$

Therefore, the time window is 80 ns.

b. Grid spacing calculation

The wavelength equation can be calculated with the following formula:

$$\lambda = \frac{c}{f\sqrt{K}}$$

$\lambda = \text{wavelength}$

$c = \text{speed of light in a vacuum}$

$f = \text{frequency}$

$K = \text{real part of permittivity}$

The GPR used dual-frequency of 200MHz and 600MHz. We will use 600MHz in this calculation. We assumed the material was dry sand with $K=4$. By applying the formula:

$$\lambda = \frac{299792458m/s}{600 * 10^6/s * \sqrt{4}}$$

$$\lambda = 0.25m$$

Therefore, the wavelength is 0.25m.

We can calculate the grid spacing using the following formulas:

$$\text{Grid spacing } \perp \leq 6 \lambda; \text{ Grid spacing } // \leq 4 \lambda$$

$$4\lambda = 0.25m * 4 = 1m$$

$$6\lambda = 0.25m * 6 = 1.5m$$

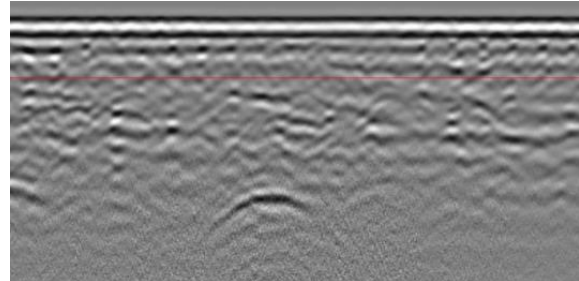
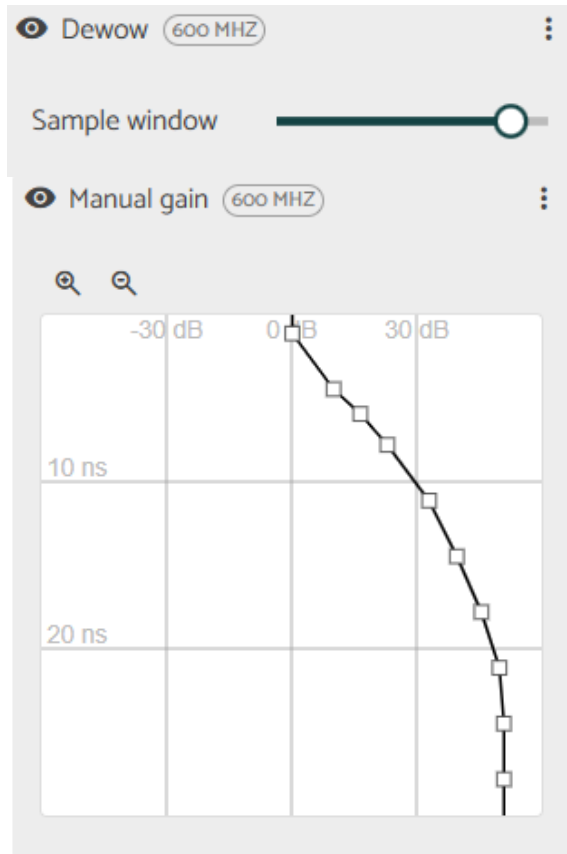
We used a grid spacing of 1m as it was within the tolerance of 4λ .

GPR signal post processing

Step 1: Signal without any processes



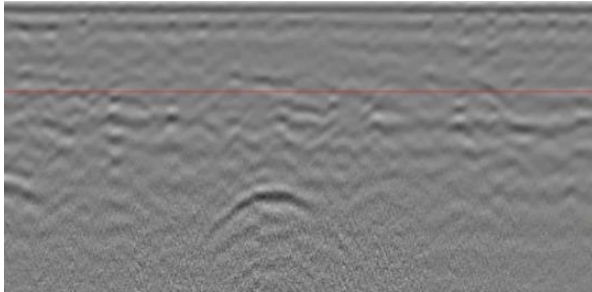
Step 2: After manual energy decay gain and dewow



The dewow process can remove the low frequency noise caused by surroundings or the antenna itself. Removing the unwanted noise can increase the contrast of the frequencies, thus improving the quality of the collected data.

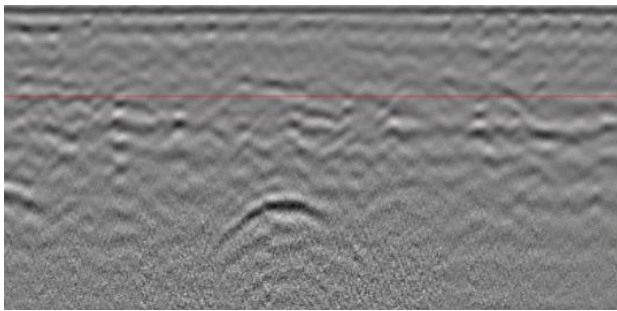
The energy decay gain process can compensate for the radar signal loss. As the EM wave passes through subsurface material, it loses energy and its signal weakens. The raw GPR signal does not account for the energy decay, therefore there is little useful data for deeper surfaces. After applying the process, the deeper signals are amplified and thus giving more information for data processing.

Step 3: After time zero correction



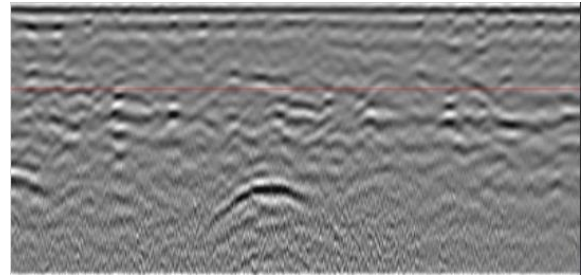
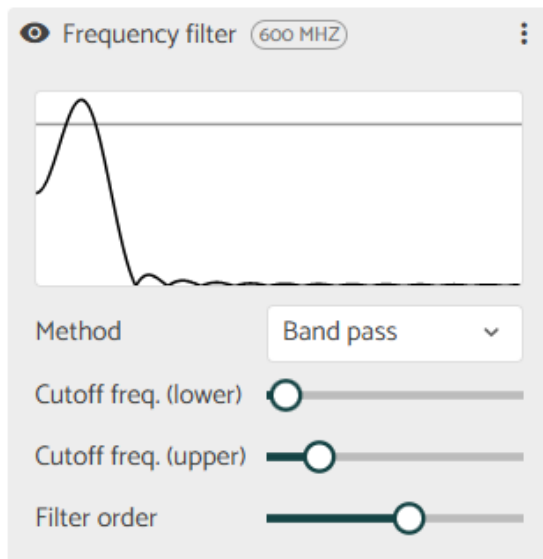
The time zero correction can account for the time the EM wave traveled through air then into the ground. Therefore, this correction must be applied to remove the time delay offset and retrieve the true depth of object.

Step 4: Normalization



The normalization process can remove signal outliers and equalize amplitudes across traces. This is because raw GPR data can have varying signal strengths depending on the location and antenna. This process rescales each trace so that the data collected remains consistent. The consistent data can also make it easier to compare the traces and identify the underground features.

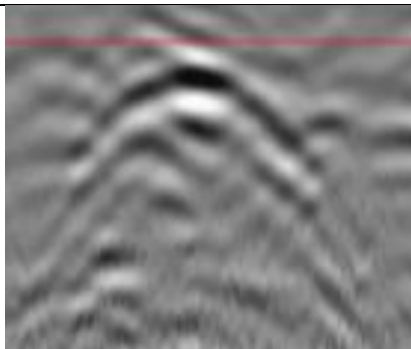
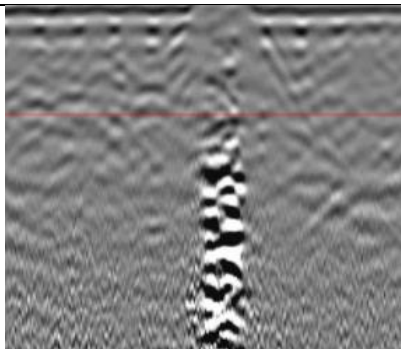
Step 5: Frequency filter



The frequency band pass filter process only keeps the data within the frequency range, which can remove frequency outliers and clean up the data. The trace can become clearer after applying the filter, allowing us to see more underground features with less noise.

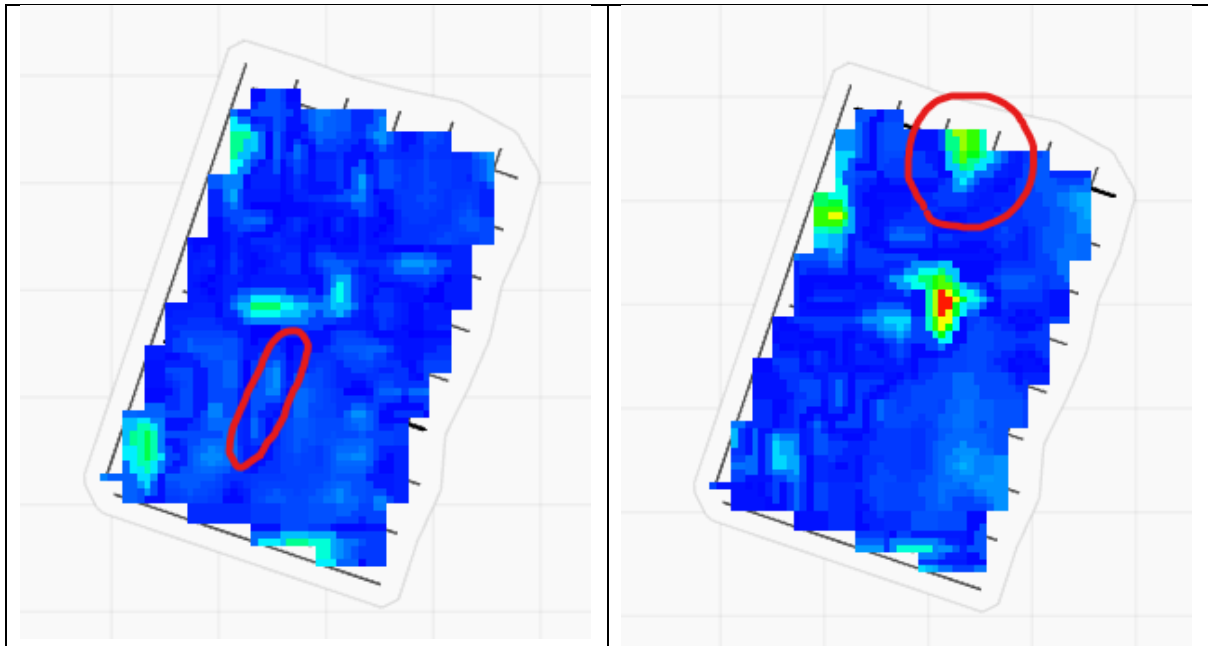
Suspected utilities

Suspected utilities in B scans

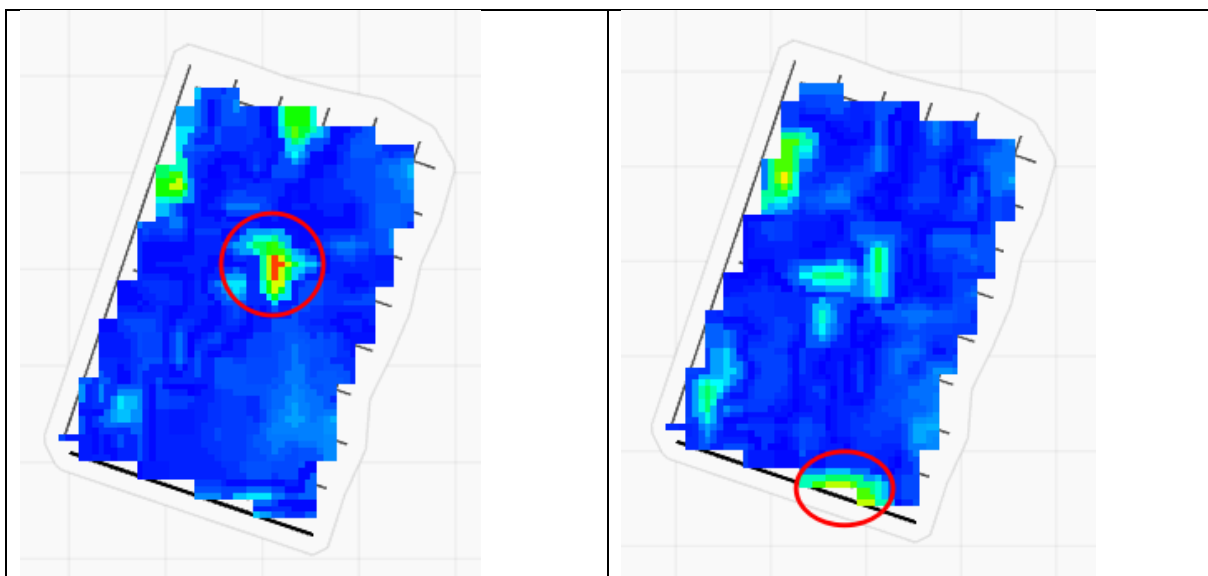
	
Pipe	Manhole

Suspected pipe locations in C scans

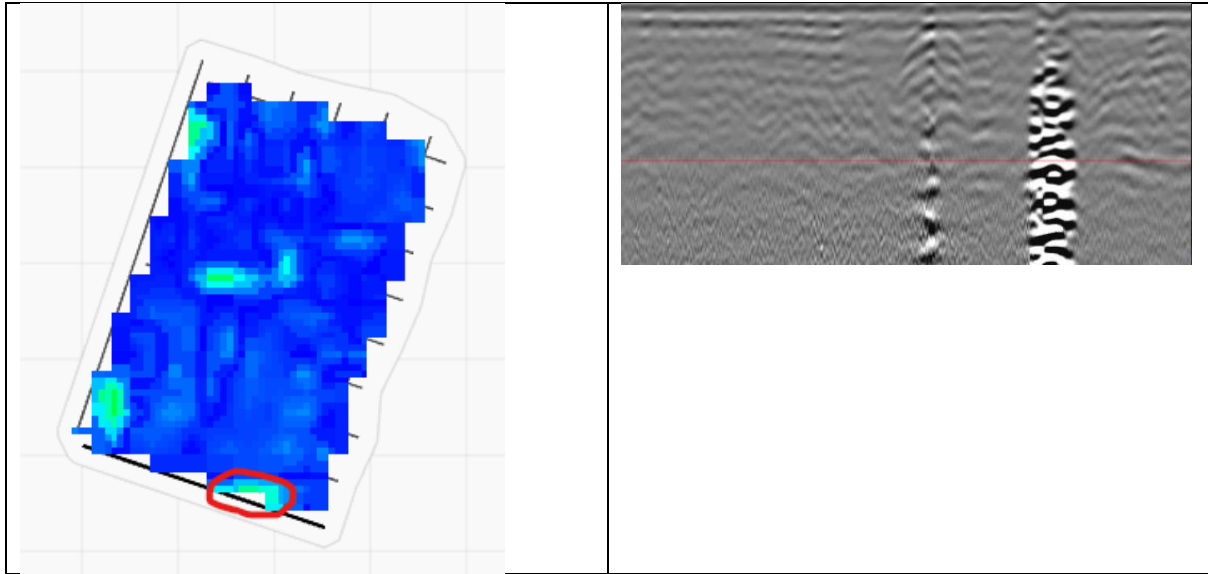
0.9m	1.06m
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Suspected manhole locations in C scans



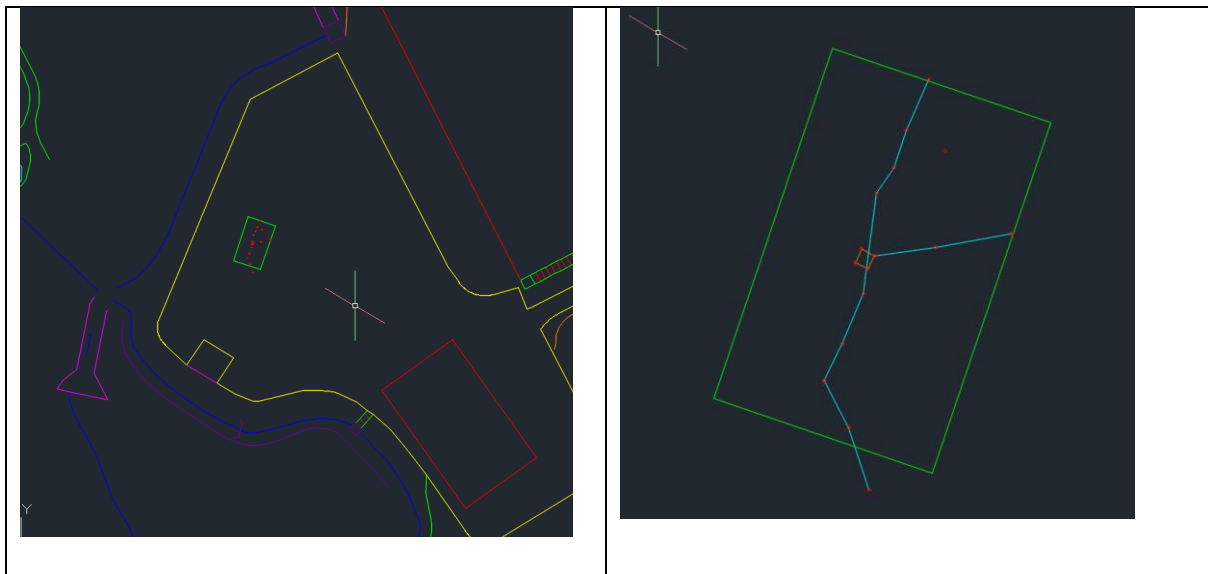
There is only 1 manhole in the southmost scan without other parabolas for suspected pipes, therefore the pipe likely connects to the manhole.



The interpretations for the manholes are from both the B scans and C scans. On most levels the suspected manhole areas have strong signals, and they also show strong continuous signals from top to bottom in both X and Y directions. Furthermore, the manhole covers were directly visible from the surface, matching the results from the scans.

However, the interpretations for the pipes are from the B scans instead of the C scans. Usually the pipes have strong and continuous signals along the alignment, however, only discrete parabolas were found without the evidence from the C scans. Therefore, the GPR survey for the pipes was **unsuccessful** and the results were **unreliable**.

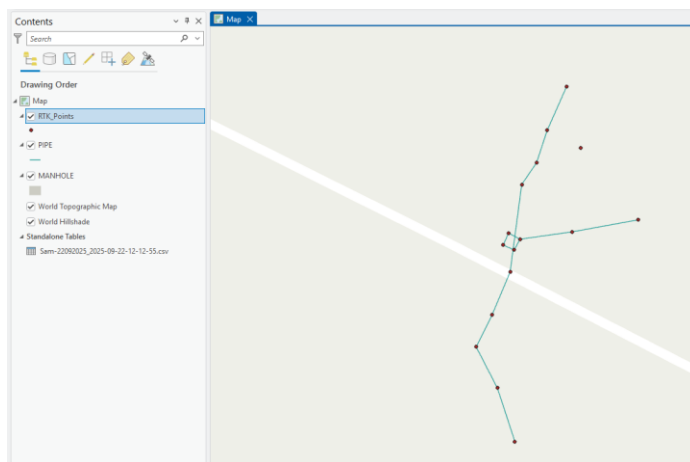
CAD Drawings



The green rectangle represents the survey area in Q leak, while the red points represent the surveyed points for PCL by RTK. There is no drawing for GPR survey due to the interpretations being unreliable.

The blue lines are drawn by connecting the red points of the RTK survey and another pipe was suspected to lie west to the GV51 valve. This result contradicts the original survey drawings, which has depicted only one pipe, connecting the SAV2 manhole in the north and the GV50 valve in the south.

ArcGIS Pro Geodatabase Drawing



Final geodatabase layer in ArcGIS Pro, including RTK points, PIPE for suspected pipes, and MANHOLE.

Quality level

As for the quality level, we believe that the suspected pipe locations reach a quality level of QL-B2, a grade for **reliable** surveying results. This is due to the strong low-high-low signals in the PCL survey and high correlation with the original survey plan. We believe that the survey plan might not contain all the underground survey data and the consistent low-high-low signals to the west of GV51 give us high confidence for a possible pipe. Although the GPR survey was unsuccessful due to the poor quality of the C scans, the overall quality is assessed by the highest among all surveying methods. Therefore, the above surveying results were reliable.

Confidence ratings at each detected UU under the frameworks of the 5 tier survey plan:

Instrument	Confidence%
Pipe & Cable Locator (PCL)	80%
Ground Penetrating Radar (GPR)	50%

RTK Survey

Methodology

The real time kinematic (RTK) surveying was being carried out after the PCL survey to find the geospatial coordinates of the detected points. In the PCL survey, after locating a point with high confidence of suspected pipes, a plastic cone was placed down at the spot, which can be later recorded by GNSS equipment. For the RTK survey, the antenna received the signal from the GNSS satellites and the mobile device can record its location. Before surveying, we also surveyed the control points A and B, to ensure that the accuracy of the equipment is within centimeter level. During the survey, GDOP should be less than 5 to ensure the positional accuracy is acceptable.

Data processing

The data was exported as an excel file, then the data can be saved as a comma separated values (CSV) file with easting, northing, and height of the points. We can import the data into AutoCAD and ArcGIS Pro and form points. The points can then be connected by 2D polylines to depict the suspected pipe locations.

Project limitations

1. Personnel

In the application of pipeline positioning technology (PCL), accurate pipeline alignment requires accumulated experience from the operators. PCL primarily uses electromagnetic signals for pipeline location; however, our group still lacks sufficient experience in equipment operation and our signal interpretation is not precise enough. For example, we might misinterpret environmental interference or noise as valid signals from the pipeline, leading to incorrect alignment.

Similarly, ground-penetrating radar (GPR) data processing also requires professional knowledge and analytical skills. Although we learned basic radar data processing and image analysis methods this semester, we still sometimes misinterpret features in radar images, posing challenges in identifying pipeline alignment and causing deviations in the target pipeline location results.

2. Equipment

GPR cannot always collect complete data. Where there is wet or clay-rich or highly conductive soils, the radar signals are strongly attenuated, which can make some subsurface features undetectable. This means that some utilities or objects may not show up in the data, leading to gaps or missing information. Furthermore, conditions of the surface, such as rough terrain or obstructions, may prohibit proper data collection, further limiting the effectiveness of GPR. GPR also struggles with differentiating between buried material types and the depth of penetration based on the frequency of the antenna used.

In contrast, Pipe and Cable Locators are limited to detecting conductive utilities like metal pipes and cables. In most cases, non-metallic utilities such as plastic or concrete pipes remain invisible unless they have tracer wires installed. PCLs may also be influenced by electromagnetic interference originating from other utilities or other sources of electricity around them, thus distorting signals and reducing the overall accuracy.

3. Materials

The difficulty of collecting subsurface utility data where promontories primarily consist of dry sand and concrete bricks is continually increasing due to the above-mentioned reasons. Since dry sand generally provides better transmission for Ground Penetrating Radar (GPR) signals than clay and moist soils, it has a good chance for a clean signal to show through. However, concrete bricks create a lot of interference due to the fact that they are solid and thick; they may reflect, scatter or absorb the radar signals. Incorrectly identified utilities may be represented poorly or distorted in the resulting images of the survey, and some may be missed completely.

4. Environment

PCL (Pipeline Penetrating Radar) pipeline detection technology is susceptible to interference from surrounding charged metal objects. Underground cables may exist within the detection area. Since we employ an active direct-connection detection method, this method may induce currents in nearby metal objects (such as fences or other underground metal pipes). The magnetic fields generated by these charged objects may interfere with the magnetic field generated by the target water pipe. If signal interference occurs, the detected peak signal location will shift, leading to positioning errors.

Ground-penetrating radar (GPR) detection results are highly sensitive to soil conditions. Different soil conditions (e.g., material composition, soil structure, and moisture) have different relative permittivity, which directly affects the penetration depth of the radar signal. This study assumes the site soil is dry sand with a relative permittivity of 5; however, this value may contain errors. These errors will affect the accuracy of GPR data depth estimation, ultimately leading to deviations in the interpretation of pipeline route depth.

Challenges

Surveying

During the fieldwork, we have encountered a couple of challenges. The first problem was that the signal resistance of the receiver was initially suboptimal, with very high Ohm value, the signal strength would be weakened. We have tried multiple locations for grounding, clamping the poles, manhole covers, and fire hydrants. Generally, the grounding location should be close to the valve and perpendicular to the alignment. However, some of these locations gave unexpectedly high resistance, and after some trial and error, we were able to find a location with signal strength high enough for PCL survey. This shows that theoretically ideal locations might not match with the environmental conditions at the survey site.

Another challenge was that during the wheel calibration process, the wheels sometimes suspend in the air due to the uneven brick ground and protruding grass. We were not confident that the calibration process was successfully carried out. To solve this problem, we changed the location and moved away from the survey area for the wheel calibration.

Data interpretation

During the data interpretation process, it was found out that the C scans yielded unreliable results for the pipes, due to lacking continuous strong reflections. Although we were able to suspect some locations of the pipes with the parabolas in the B scans, the points were too discrete and sparse to confidently locate the pipes. In the GPR survey, we used a 1m grid spacing, which was the minimum required for 4λ . For the survey recommendations, a smaller grid spacing should be used, and the survey accuracy will likely increase.

Conclusion

This underground utility survey at survey site 3, Q-Leak has successfully identified two possible pipe alignments. The primary alignment from SAV2 to GV50 was confirmed, while a secondary, previously uncharted alignment extending westwards from GV51 was discovered. The PCL survey coupled with RTK positioning proved to be a highly reliable method, providing strong evidence for these alignments through consistent low-high-low signals and precise geospatial referencing.

In contrast, the GPR survey yielded mixed results. It was successful in clearly identifying the locations of manholes, which were corroborated by surface evidence and showed strong, continuous reflections in both B-scan and C-scan data. However, the GPR survey was unsuccessful in providing reliable evidence for pipe locations. The C-scan data lacked the continuous linear features necessary for confident interpretation, and the parabolic hyperbolas in the B-scans were too discrete. This highlights a critical limitation of GPR in certain site conditions, potentially due to soil composition, signal attenuation, or the 1m grid spacing being at the theoretical minimum, which may have been insufficient to resolve smaller or deeper utilities.

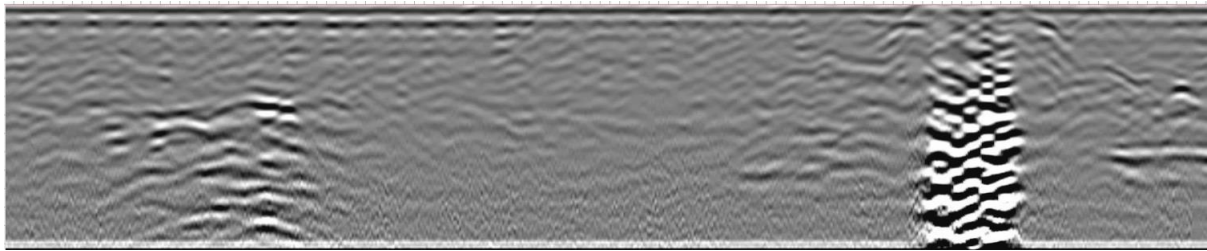
Multiple challenges were encountered, including optimizing PCL signal strength through iterative grounding and addressing practical issues with GPR wheel calibration on uneven terrain. These experiences provided valuable lessons in problem-solving and equipment handling during fieldwork.

Overall, this project underscores the complementary nature of geophysical survey methods. The failure of one method (GPR for pipes) was compensated by the success of another (PCL), demonstrating the necessity of a multi-technique approach for robust and reliable utility mapping. The discovery of a potential unrecorded pipe west of GV51 emphasizes the value of such surveys in updating and validating existing utility records.

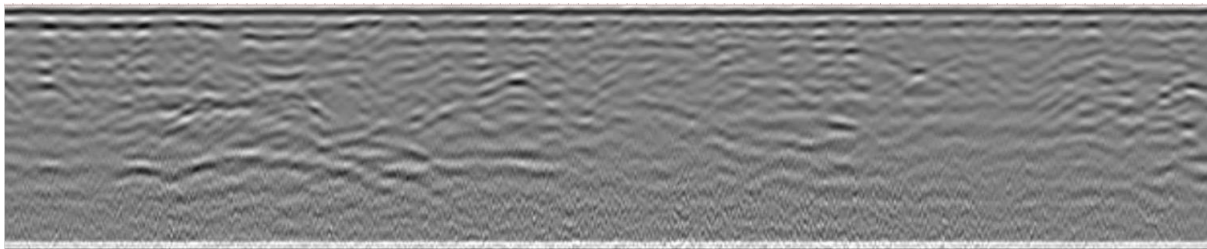
For future work, we recommend:

- Employing a denser GPR survey grid (e.g., 0.5m spacing) refined wheel calibration and possibly a different antenna frequency could improve horizontal resolution and penetration, while repeated PCL with alternative grounding points and frequencies would help reduce signal interference and increase confidence in the final utility plan.
- Integrating the data into a unified GIS platform for all future surveys to facilitate better visualization, analysis, and long-term asset management.

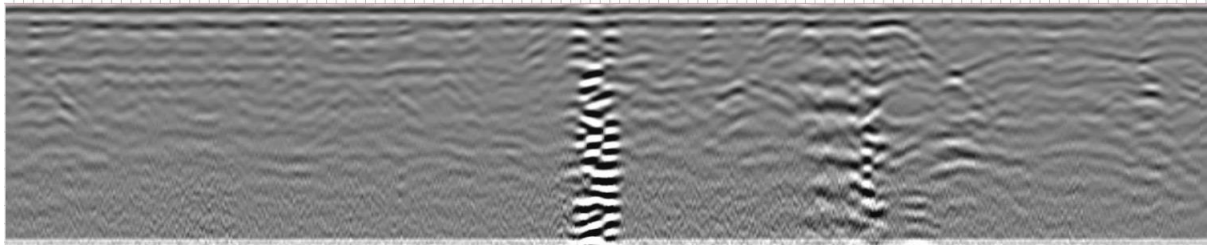
Appendix A – B scans



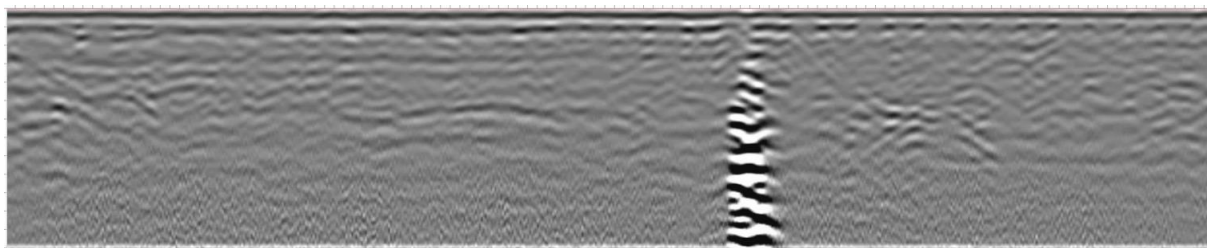
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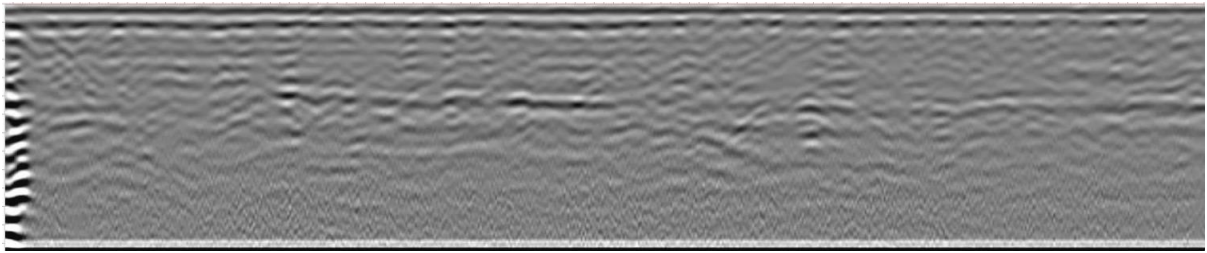
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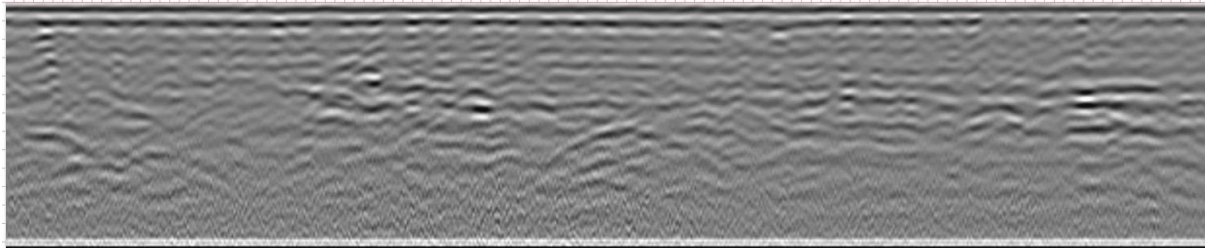
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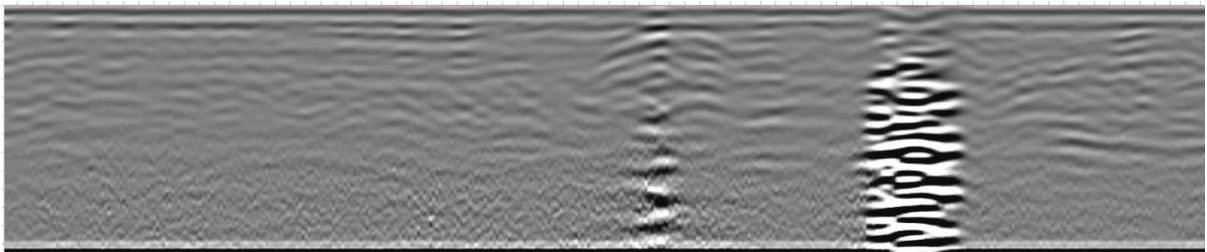
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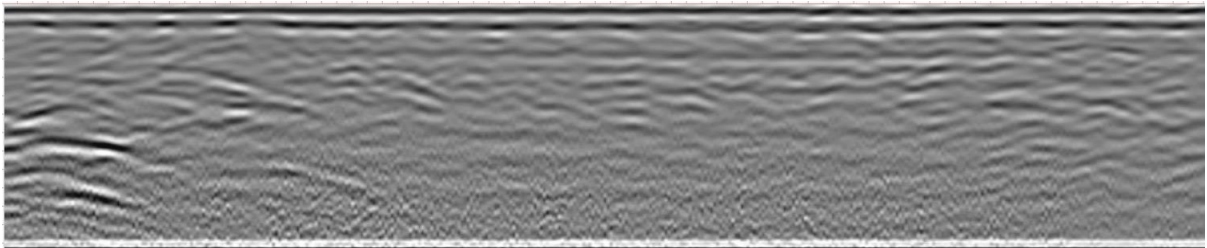
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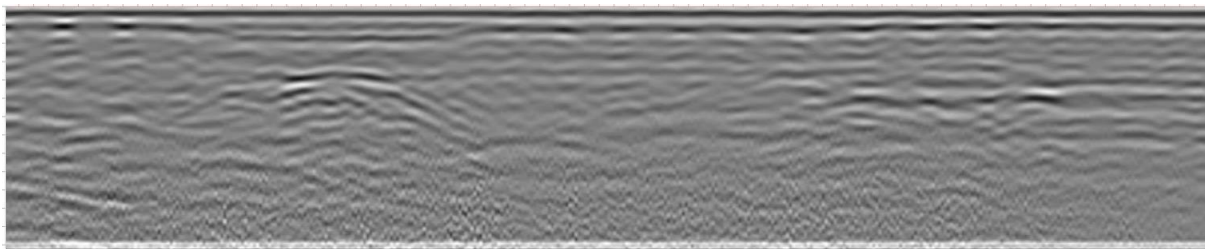
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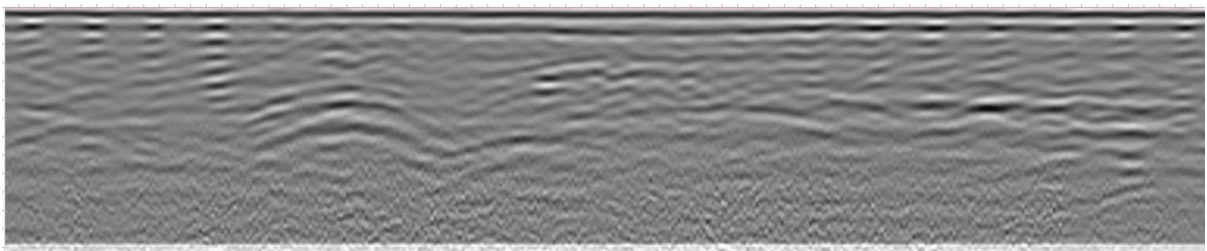
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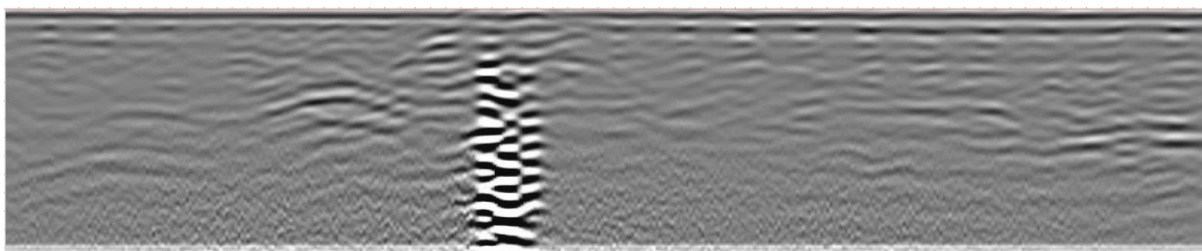
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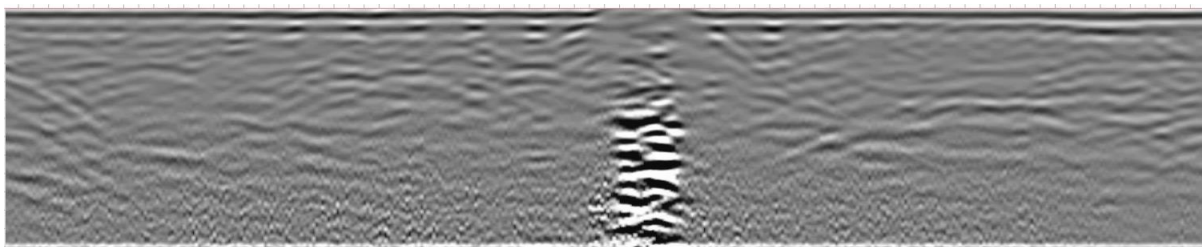
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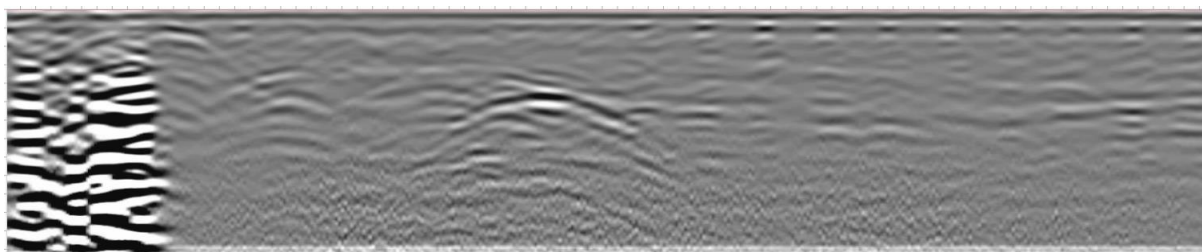
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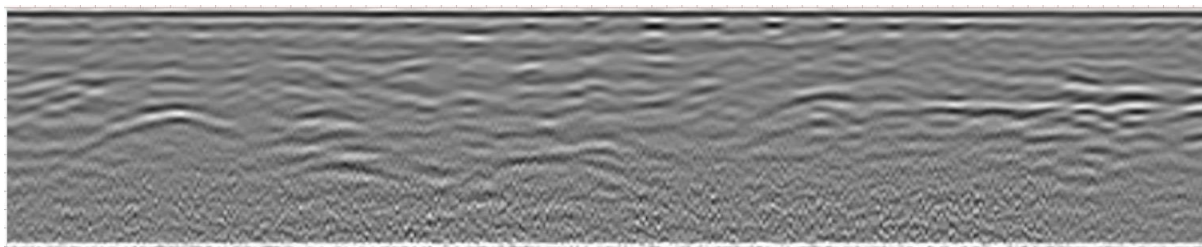
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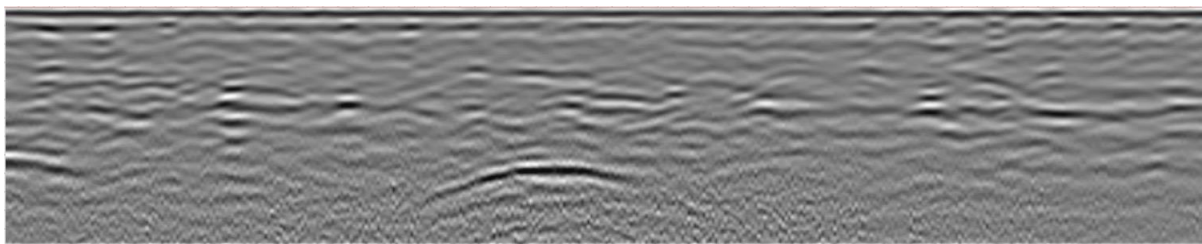
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Appendix B – Site Photos

